

Technical Document #218: Retrieval Augmented Generation - Comprehensive Overview

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Vector databases store dense embeddings for semantic search. Pinecone, Weaviate, and ChromaDB are popular vector database solutions.

Semantic search finds documents based on meaning rather than exact keywords. It uses embedding similarity to match queries with relevant content.

Chunking splits documents into smaller segments for embedding. Optimal chunk size balances context retention with retrieval precision.

Hallucination detection identifies when LLMs generate factually incorrect information. It is critical for maintaining trust in RAG systems.

Re-ranking models score retrieved documents for relevance to the query. They improve the quality of context provided to the generator.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Key Takeaways:

- Hybrid search combines keyword-based and semantic search.
- Context window optimization selects the most relevant information to fit within the LLM's context limit.
- Semantic search finds documents based on meaning rather than exact keywords.